1 Architecture Trade Study

Before selecting a constellation topology (section 2) or deriving throughput scaling laws (section 3), a higher-level question must be answered: what class of architecture can plausibly deliver gigabit-per-second *bidirectional* bandwidth between Earth and Mars? A Mars settlement requires high-rate data flow in both directions—Earth-to-Mars for software updates, AI model weights, and entertainment; Mars-to-Earth for telemetry, scientific data, and video. Any viable architecture must support symmetric or near-symmetric throughput. We evaluate three candidates: direct point-to-point links (including Lagrange-point relays), synthetic-aperture arrays, and many-satellite relay chains.

1.1 Direct Links and Lagrange-Point Relays

The simplest architecture places one or a few high-capability relay spacecraft on the Earth–Mars line and pushes as much power through a single optical link as technology allows. NASA's Deep Space Optical Communications (DSOC) experiment aboard the Psyche spacecraft provides the most relevant data point. DSOC achieved a peak downlink rate of 267 Mbps from a distance of 31 Mkm [**?**], using a 4 W flight laser transmitter with a 22 cm aperture and the 5 m Hale Telescope at Palomar as the ground receiver. As Psyche receded, the inverse-square law took hold: at 390 Mkm (roughly Mars apoapsis distance from Earth), the maximum rate fell to 8.3 Mbps [**?**].

These results confirm both the promise and the fundamental limitation of direct links. Notably, DSOC is a *unidirectional* demonstration: the 267 Mbps figure is the downlink (Mars-to-Earth) rate, achieved by pairing a small flight transmitter with an enormous ground receiver. The uplink (Earth-to-Mars) used a separate 5 kW ground laser solely for beacon tracking and synchronisation, not for high-rate data. A bidirectional direct-link system would need DSOC-class terminals at *both* ends—including a large-aperture receiver at Mars, where no such infrastructure exists.

At close range, free-space optical communication is extraordinarily capable. But throughput falls as $1/d^2$: doubling the distance quarters the data rate. Extrapolating from DSOC's demonstrated performance, a single equivalent terminal at the average Earth–Mars separation of $D_{\mathrm{EM}} \approx 225$ Mkm would deliver roughly 6 Mbps—adequate for robotic telemetry, but three orders of magnitude below the 1 Gbps target needed for human settlement support (medical imaging, software updates, AI model synchronisation, real-time video). Reaching 1 Gbps on a single direct link at Mars-average distance would require increasing the K parameter by a factor of ~ 170 over DSOC—equivalent to, say, a 250 W laser with a 2 m flight aperture feeding a 50 m ground receiver. No such system is on any near-term roadmap.

Placing relays at the Sun–Earth or Sun–Mars Lagrange points (L4, L5) addresses a different problem: solar conjunction, the periodic geometry in which the Sun blocks the direct Earth–Mars line. L-point relays can route around the Sun, maintaining link availability during conjunction. However, L-point relays

are still at interplanetary distances from one or both planets ($\sim$150 Mkm from Earth to Sun–Mars L4) and therefore suffer the same inverse-square throughput penalty. They are best understood as a supplement to whatever primary architecture is chosen—useful for conjunction mitigation, but not a solution to the throughput problem.

1.2 Synthetic-Aperture Arrays

A second approach attempts to defeat the inverse-square law not by shortening the link but by scaling up the effective antenna area. Many small satellites in orbit around each planet could collectively form a synthetic aperture far larger than any individual telescope, boosting the link budget without requiring a single monolithic receiver.

A crucial asymmetry simplifies the transmit side of the problem. Transmission does not require coherent combining: a single satellite with a sufficiently powerful laser can illuminate the far planet, and the beam divergence is set by its physical aperture alone. No phase coordination between multiple transmitters is needed (though coherent transmit arrays are possible in principle, they are not required). The hard problem is entirely on the *receive* side, where the faint signal arriving from interplanetary distance must be collected by many apertures and coherently combined into a single data stream.

For the Mars-to-Earth direction, a single transmitter at Mars sends a laser beam toward Earth, where a constellation of N_E satellites in Earth orbit each collect a fraction of the arriving photons. If their received signals are coherently combined—phase-aligned to a fraction of the optical wavelength and summed—the effective receive aperture is $A_{\text{eff}} = N_E\, A_0$, where A_0 is the aperture of each element. The Earth-to-Mars direction mirrors this: a single ground-based or orbital transmitter at Earth sends toward Mars, and a constellation of N_M satellites in Mars orbit forms the synthetic receive array.

Bidirectional communication therefore requires a coherent receive array at *each* planet. A single transmitter at each end suffices, but both the Earth-orbit and Mars-orbit constellations must independently solve the coherent combining problem.

To reach 1 Gbps at Mars-average distance starting from DSOC-class parameters, the receive end needs an effective aperture gain of ~ 170 over the 5 m Hale Telescope, implying a synthetic aperture equivalent to a ~ 65 m dish—achievable with $\sim$170 satellites each carrying a 5 m telescope, coherently phase-locked across baselines of hundreds of kilometres. This is a modest constellation by modern standards (Starlink has $> 6{,}000$ satellites), but the per-satellite complexity is in a different class entirely.

The technical barrier is coherent combining at optical wavelengths over orbital baselines. Phase synchronisation must be maintained to ~ 100 nm (a fraction of the 1550 nm carrier wavelength) across kilometre-scale separations, in real time, while each satellite independently acquires and tracks the incoming signal. Multi-aperture coherent combining has been demonstrated in laboratory settings and over short atmospheric paths [**?**], but never at orbital baselines or

interplanetary distances. The gap between demonstrated and required capability is not incremental—it is a qualitative leap in precision metrology, distributed clock synchronisation, and wavefront control.

Even if coherent combining were solved at one planet, the bidirectional requirement means it must be solved *twice*: once in Earth orbit and once in Mars orbit, in different gravitational and thermal environments, with different orbital dynamics, and with a light-travel delay that precludes real-time coordination between the two arrays.

The architecture also has a structural scaling disadvantage. It improves the *aperture* but not the *distance*: the link still traverses the full Earth–Mars gap, so the received photon flux per unit area remains fixed by the inverse-square law. The synthetic aperture merely collects more of those photons. To double the data rate, one must double the total collecting area—a linear scaling that competes poorly with the super-linear ($S^{3/2}$) scaling of a relay chain, as we show in section 3.5.

In summary, the synthetic-aperture approach requires two constellations of specialised satellites (one at each planet), both solving the same unsolved engineering problem (orbital coherent combining at optical wavelengths) in different environments, plus a single high-power transmitter at each end. The relay chain, by contrast, uses a single design of simple, identical satellites with two laser terminals each, is inherently bidirectional, requires no coherent combining, and scales super-linearly.

1.3 Many Inexpensive Relay Satellites

The third approach applies the Starlink paradigm to interplanetary space: instead of one powerful link across the full Earth–Mars gap, deploy a chain of many inexpensive relay satellites at short intervals, each forwarding data to its neighbour. Because each satellite both receives from one neighbour and transmits to the next, the relay chain is *inherently bidirectional*—the same hardware and the same inter-satellite links carry data in both directions simultaneously, with no asymmetry between Earth-to-Mars and Mars-to-Earth throughput.

1.3.1 The Starlink Baseline

The technological foundation already exists. Starlink's inter-satellite laser links operate at 100 Gbps over distances up to 5,400 km, with over 9,000 space lasers in service and $> 99\%$ link uptime across the fleet [**?**]. These are mass-produced terminals—manufactured at a rate exceeding 40 units per week, at costs driven well below \$1 M per satellite by vertical integration and iterative manufacturing. The Starlink laser link is the empirical anchor for this paper's baseline: $C_0 = 100\,\text{Gbps}$ at reference distance $d_0 = 3{,}000\,\text{km}$, giving $K = C_0\, d_0^2 = 9 \times 10^8\,\text{Gbps}\cdot\text{km}^2$ (see section 3.1).

A MarsLink relay satellite is *simpler* than a Starlink unit: it needs no user-beam phased array, no RF inter-satellite links, no ground-facing Ka/Ku-band antennas—just two laser terminals (one inward, one outward) and a solar panel.

The per-unit cost could plausibly be comparable to or lower than a Starlink satellite.

1.3.2 The Technology Improvement Factor

Throughout this paper we parameterise future technology improvements through a multiplicative factor f applied to the baseline capacity: $K = f\, C_0\, d_0^2$. Several independent avenues can increase f, each with different cost and difficulty profiles:

- **Increased laser power.** Starlink terminals currently operate at modest power levels ($\sim$1–2 W optical). Scaling to 10–50 W is well within the state of the art for space-qualified fibre lasers, yielding $f = 5$–25 from power alone.
- **Larger aperture.** Increasing the transmit/receive aperture from the current $\sim$10 cm class to 20–30 cm improves the link budget as the square of the aperture ratio, giving $f = 4$–9.
- **Better modulation and coding.** Higher-order modulation formats (e.g., 16-QAM, 64-QAM) and improved forward error correction can extract 2–4$\times$ more bits per photon, contributing $f = 2$–4.
- **Receiver sensitivity.** Photon-counting detectors (superconducting nanowire single-photon detectors, or SNSPDs) offer $\sim 10\times$ improvement in sensitivity over conventional avalanche photodiodes, though at higher cost and complexity.
- **Wavelength-division multiplexing (WDM).** Running multiple wavelength channels on the same aperture scales capacity linearly with channel count, at the cost of additional laser and receiver hardware per channel.

These improvements compound multiplicatively: a relay satellite with a 30 W laser (15$\times$), a 25 cm aperture (6$\times$ over 10 cm), and improved coding (2$\times$) would achieve $f \approx 180$, delivering $\sim$100 Gbps per link at $\sim$40,000 km—comfortably spanning typical inter-ring gaps in the constellation geometries analysed in section 3. In practice, the combination yielding the lowest marginal cost increase will be preferred, balanced against the total satellite count: investing more in each satellite (higher f) reduces the number needed, while keeping satellites cheap (low f) requires more of them. The optimal trade-off between per-satellite capability and constellation size is a manufacturing economics question, not a physics question—and SpaceX's vertically integrated production model is uniquely positioned to navigate it.

1.3.3 The Inverse-Square Advantage

The key insight is how violently the inverse-square law favours many short links over few long ones. Compare two ways to spend a budget of S satellites:

- **Single direct link** at distance $D_{\mathrm{EM}} \approx 225$ Mkm (average Earth–Mars): throughput $\sim K/D_{\mathrm{EM}}^2 \approx 0.02$ kbps ($f = 1$).
- **Relay chain** of S satellites spanning D_{EM}, with inter-satellite gap $D_{\mathrm{EM}}/(S+1)$: the bottleneck link capacity is $K\left(\frac{S+1}{D_{\mathrm{EM}}}\right)^2$, which for $S = 100$ already exceeds the direct link by a factor of $\sim$10,000.

Every relay satellite added shortens the inter-satellite gap, and the inverse-square law amplifies this quadratically. As derived in section 3.5, when satellites are distributed optimally across multiple parallel rings rather than a single chain, the aggregate throughput scales as $S^{3/2}$—super-linearly with constellation size.

1.3.4 The Central Question

The relay architecture transforms the interplanetary communication problem from a physics problem (how to push a signal across 225 Mkm) into an engineering and manufacturing problem: *how many satellites, arranged where, for what total performance?* Starship's payload capacity to heliocentric orbit ($\sim$100–150 tonnes to trans-Mars injection) enables deployment of hundreds of relay satellites per launch, amortising launch costs across the constellation. The remaining design variables are the number of satellites S, the number of relay rings R, the number of routes per ring $N = S/R$, and the technology factor f. The throughput analysis of section 3 derives the optimal allocation and the resulting scaling laws.

1.4 Architecture Comparison

Table 1 compares the three architectures across four dimensions: cost, reliability, redundancy, and time-to-market feasibility.

Cost. A direct-link architecture is the cheapest to deploy in one direction—a single DSOC-class terminal costs on the order of \$50–100 M including spacecraft bus and launch—but delivers only single-digit Mbps at Mars distances, so the cost *per Gbps* is effectively infinite (the target is unreachable). Making the link bidirectional roughly doubles the cost, since a large-aperture receiver must be deployed at Mars as well as at Earth. Adding Lagrange relays multiplies the cost further without improving throughput. A synthetic-aperture system needs a coherent receive array at each planet for bidirectional operation: $\sim$170 satellites per array, each carrying a 5 m-class telescope with sub-wavelength metrology, at an estimated \$50–100 M per unit (comparable to a space observatory), plus a single high-power transmitter at each end. The total for both receive arrays plus both transmitters is on the order of \$20–40 B—before the coherent-combining problem is solved. A relay constellation of $S = 10{,}000$ satellites at \$5.4 M per unit plus \$1 M per unit in amortised Starship launch costs totals $\sim$\$64 B, but delivers $\sim$ 30 Mbps bidirectionally at $f = 1$ (worst case), falling to $\sim$\$6 B for $S = 1{,}000$ satellites at lower throughput. Crucially, the relay cost per Gbps

decreases with scale due to the $S^{3/2}$ throughput scaling (section 3.5), and the same hardware carries traffic in both directions at no additional cost.

Reliability and redundancy. A direct-link system has zero redundancy in each direction: one spacecraft failure means total loss of service, and the replacement timeline is measured in years (manufacture, launch window, cruise). For bidirectional operation, the system has *two* independent single points of failure (the Earth-end and Mars-end terminals). Lagrange relays add route diversity for conjunction avoidance but not for hardware failure. A synthetic-aperture array degrades gracefully on the receive side—losing one of 170 elements reduces effective aperture by $< 1\%$—but the coherent-combining control system is itself a single point of failure at each planet, and the single transmitter at each end is also a single point of failure. The bidirectional system thus has four critical subsystems (two receive arrays, two transmitters), any one of which can take down one direction of traffic. A relay constellation is inherently redundant at the architecture level: in the parallel relay topology (section 2.3), losing one satellite costs one of N parallel routes, a throughput reduction of $1/N$ in *both* directions simultaneously. No single component failure can cause a total outage in either direction.

Time to market. The direct-link approach is partially deployable today—DSOC has demonstrated the downlink—but a bidirectional 1 Gbps system requires a generational leap in both laser power and receiver infrastructure at Mars (estimated 15–20 years). Synthetic aperture at optical wavelengths requires solving space-based coherent combining over orbital baselines at both planets, a problem with no demonstrated solution and no funded development programme; a conservative estimate is 20+ years. A relay constellation requires no new physics and no undemonstrated technology—only the scaling of existing Starlink-class manufacturing to a simpler satellite design, plus Starship operational readiness for heliocentric deployment. Because the same relay hardware carries traffic in both directions, there is no separate "uplink programme" to develop. SpaceX's current production trajectory and Starship development timeline suggest initial operational capability within 5–10 years.

Figure 1 illustrates the core argument. A single DSOC-class link drops below 1 Mbps at typical Mars distances, while a relay chain maintains high per-link capacity by keeping every hop short.

We adopt the many-satellite relay architecture for the remainder of this paper. The next question—how to arrange those satellites—is addressed in section 2.

Table 1: Architecture comparison for achieving 1 Gbps bidirectional Earth–Mars bandwidth.

Criterion	**Direct / L-point**	**Synthetic aperture**	**Relay chain**
Bidirectional design	Separate TX/RX per end	2 receive arrays + 2 TX	Same hardware, both d
Peak demonstrated rate	267 Mbps (downlink only)	N/A (unbuilt)	N/A (unbuilt)
Bidir. rate at Mars avg.	∼6 Mbps (each dir.)	∼1 Gbps (if coherent)	∼30 Mbps–1 Gbps
Throughput scaling	$1/d^2$ (fixed)	Linear in array size	$S^{3/2}$
Est. bidir. system cost	\$0.2–1 B	\$20–40 B	\$6–64 B
Cost per Gbps trend	∞ (unreachable)	Decreasing (linear)	Decreasing (super-linea
Redundancy	None (per direction)	Graceful (aperture only)	Graceful (per-route, bo
Single points of failure	1 per direction	4 (2 arrays + 2 TX)	None
Key unsolved technology	None (but insufficient)	Orbital coherent combining	None
Time to bidir. 1 Gbps	15–20 yr	20+ yr	5–10 yr

2 Topology Comparison: Serial, Hybrid, and Parallel

Having established in section 1 that a constellation of many inexpensive relay satellites is the preferred architecture, the next design decision is how to *arrange* those satellites. The topology determines how data flows from Earth to Mars: how many independent paths exist, which link is the bottleneck, and how the constellation responds to individual satellite failures. We evaluate three candidate topologies—serial, hybrid, and parallel—against four criteria: aggregate throughput scaling, redundancy, failure tolerance, and hardware complexity per satellite. Figure 2 illustrates the three topologies schematically.

2.1 Serial Relay (Single Transfer Orbit)

The most intuitive relay topology uses three orbital rings. A circular *Earth ring* of N_E satellites orbits at heliocentric radius r_{E}. A circular *Mars ring* of N_M satellites orbits at r_{M}. Between them, a single *transfer ring* of N_H satellites follows an elliptical Hohmann-like orbit whose perihelion coincides with r_{E} and whose aphelion coincides with r_{M}. Data flows in series: ground stations hand off to the Earth ring, the Earth ring relays to the transfer ring near perihelion, the transfer ring relays to the Mars ring near aphelion, and the Mars ring delivers to the surface.

Each satellite on the Earth and Mars rings carries two laser terminals (circumferential, to maintain the ring) plus one inter-ring terminal pointing toward the transfer ring. Each transfer-ring satellite carries two circumferential terminals plus two inter-ring terminals (one Earth-facing, one Mars-facing), for a total of four.

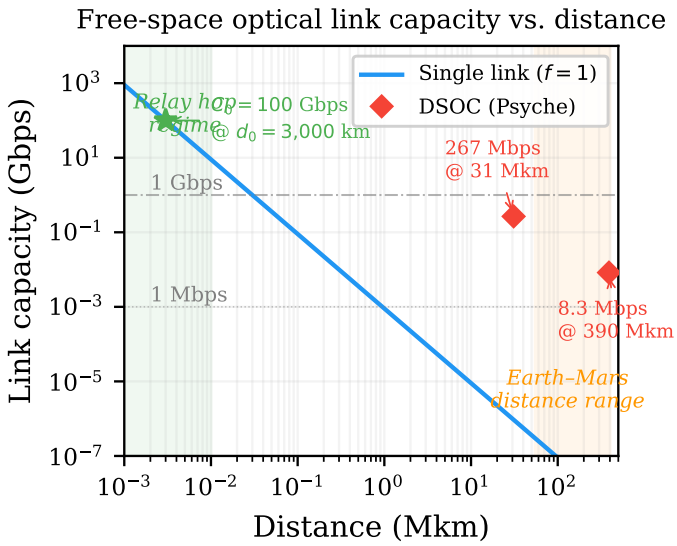

Figure 1: Free-space optical link capacity vs. distance for a single DSOC-class terminal ($f = 1$, $K = 9 \times 10^8\,\mathrm{Gbps} \cdot \mathrm{km}^2$). The DSOC data points are from NASA's Psyche mission [**?**]. The shaded region marks the Earth–Mars distance range. A relay chain keeps every hop in the high-capacity regime (left of the plot), avoiding the interplanetary penalty entirely.

The bottleneck is the inter-ring link. At perihelion, transfer-ring satellites are co-orbital with the Earth ring, and the link distance is set by the angular separation between the nearest satellites on each ring—of order $\pi r_{\mathrm{E}}/N_E$ for the Earth ring. At aphelion, the same applies at Mars: the link distance is of order $\pi r_{\mathrm{M}}/N_M$. Between these extremes, transfer-ring satellites spend the majority of their orbital period at intermediate heliocentric distances where they are far from both planetary rings. During cruise, the link to the nearest planetary ring spans a distance of order $D_{\mathrm{EM}}/2 \approx 39$ Mkm, and the inverse-square law reduces per-link capacity to

$$C_{\mathrm{cruise}} \sim \frac{K}{(D_{\mathrm{EM}}/2)^2} = \frac{4K}{D_{\mathrm{EM}}^2} \tag{1}$$

For the baseline parameters ($K = 9 \times 10^8\,\mathrm{Gbps} \cdot \mathrm{km}^2$, $D_{\mathrm{EM}} = 78 \times 10^6\,\mathrm{km}$), this gives approximately 0.6 kbps—far below any useful threshold. The transfer ring is only effective when its satellites are near perihelion or aphelion; the rest of the orbit is a dead zone.

Because the interplanetary segment passes through a single transfer ring, the topology has only one data path between Earth and Mars at any given time. This creates a single point of failure: the loss of the active transfer-ring relay severs the link entirely. The Earth and Mars rings provide local redundancy (a

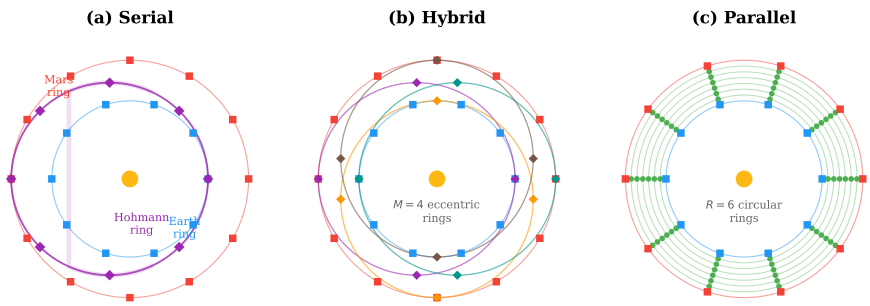

Figure 2: Schematic comparison of the three relay topologies (top-down heliocentric view). **(a)** Serial: a single transfer-orbit ring connects the Earth and Mars circular rings at perihelion and aphelion. **(b)** Hybrid: M transfer-orbit rings with staggered arguments of perihelion improve coverage. **(c)** Parallel: R concentric circular rings create N independent radial routes.

failed satellite on either circular ring can be bypassed via its neighbours), but this is irrelevant if the interplanetary hop is broken.

Throughput does not scale well with constellation size. Adding satellites to the Earth or Mars rings improves local connectivity but does nothing for the interplanetary bottleneck. Adding satellites to the transfer ring increases the number of potential relay nodes near perihelion and aphelion, but only one is active at a time (the one closest to the handoff point), so throughput remains limited to a single inter-ring link.

2.2 Hybrid Relay (Multiple Transfer Orbits)

The hybrid topology retains the Earth and Mars circular rings of the serial configuration but replaces the single transfer ring with M transfer rings, each on its own Hohmann-like orbit. All M orbits share the same semi-major axis and eccentricity (perihelion at r_{E}, aphelion at r_{M}) but have their arguments of perihelion equally spaced at intervals of $360°/M$. This angular staggering ensures that, at any instant, the M transfer rings are distributed around the Sun at different phases of their orbits, so that at least some satellites are always near one of the two planetary rings.

The motivation is to address the dead-zone problem of the serial topology. With M transfer rings, the probability that *some* transfer satellite is near perihelion or aphelion at any given time increases with M. For large enough M, the constellation can maintain near-continuous handoff between the Earth ring and whichever transfer ring is currently closest, and likewise between the Mars ring and its nearest transfer neighbour.

However, the fundamental throughput bottleneck remains. Each transfer ring still traverses the full Earth–Mars gap, and a satellite on any such ring spends the majority of its orbital period at intermediate heliocentric distances

where the link to either planetary ring spans tens of millions of kilometres. The inverse-square penalty during cruise (eq. (1)) applies identically to every transfer ring. Staggering the arguments of perihelion improves *availability*—the fraction of time that at least one usable relay exists—but does not improve the *throughput* of any individual relay during cruise.

The hybrid topology does gain a measure of redundancy over the serial configuration. With M transfer rings, the loss of one ring still leaves $M - 1$ alternative interplanetary paths. But each path independently suffers the same cruise-phase throughput floor, so the aggregate capacity scales at best linearly with M—and only during the favourable orbital phases when a transfer satellite is near a planetary ring. During cruise, additional transfer rings add satellites that contribute negligible throughput.

Hardware complexity also increases. Each transfer-ring satellite requires four laser terminals (two circumferential, one Earth-facing, one Mars-facing), and the pointing geometry changes continuously along the eccentric orbit. Satellites on the Earth and Mars rings need additional terminals to maintain links with whichever transfer ring is currently nearby. The total terminal count per satellite is 3–4, compared with 2 for the parallel topology.

Figure 3 quantifies the dead-zone problem. For a single transfer ring ($M = 1$), the best available inter-ring link capacity drops by over eight orders of magnitude between the perihelion/aphelion peaks and the mid-orbit trough. Adding transfer rings ($M = 3, 6$) raises the envelope by introducing peaks at intermediate phases, but the capacity between peaks remains far below the constant throughput achievable with the parallel topology.

2.3 Parallel Relay (Concentric Circular Orbits)

The parallel topology distributes the S satellites across R concentric heliocentric circular orbits ("rings") between Earth and Mars. Each ring contains $N = S/R$ satellites evenly spaced around its circumference. Every satellite carries exactly two laser terminals: one pointing radially inward to the nearest satellite on the adjacent inner ring, and one pointing radially outward to the nearest satellite on the adjacent outer ring. There are no circumferential (intra-ring) links.

This arrangement creates N independent parallel routes from Earth to Mars, each consisting of a radial chain of R inter-ring hops. The throughput of each route is limited by its bottleneck link (the outermost inter-ring gap, as derived in section 3.3), and the total constellation throughput is the sum over all N routes:

$$T_{\text{parallel}} \;=\; N \cdot \frac{K}{d_{\max}^2} \tag{2}$$

As shown in section 3.5, optimising the distribution between rings and routes yields $T_{\text{opt}} \propto S^{3/2}$. This super-linear scaling is the parallel topology's defining advantage: every satellite added improves both the per-route capacity (by shortening inter-ring distances) and the aggregate capacity (by adding routes).

Redundancy is intrinsic. The loss of a single satellite removes one route from one ring but leaves all other $N - 1$ routes intact. Throughput degrades

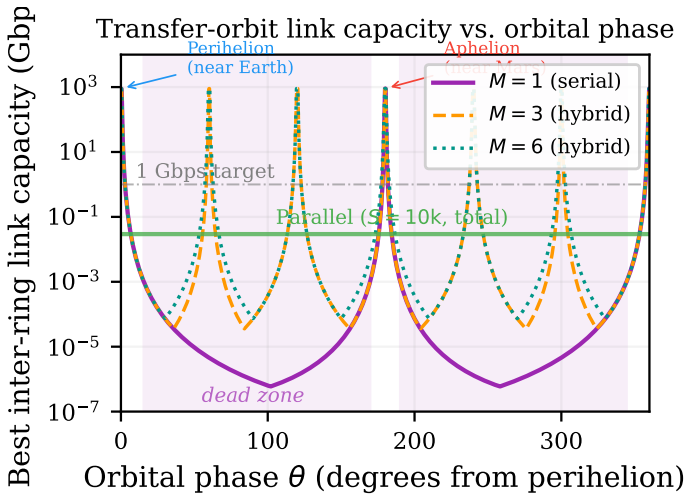

Figure 3: Best available inter-ring link capacity as a function of orbital phase for the serial ($M = 1$) and hybrid ($M = 3, 6$) topologies. The horizontal green line shows the total throughput of a parallel constellation with $S = 10{,}000$ satellites at optimal ring allocation. Baseline parameters: $f = 1$, $K = 9 \times 10^8$ Gbps$\cdot$km^2.

gracefully—by a fraction $1/N$—rather than catastrophically. For a constellation with $N = 100$ routes per ring, losing one satellite reduces total throughput by 1%.

The hardware requirement is minimal: two laser terminals per satellite, the fewest of any topology considered. The two terminals point radially (inward and outward), and each satellite participates in exactly one route. This radial-only pointing simplifies acquisition and tracking, since adjacent rings are separated by a fixed, predictable gap.

Scalability follows naturally. Adding satellites to existing rings increases N and therefore the number of parallel routes. Adding new rings between existing ones increases R, shortening inter-ring distances and boosting per-route capacity. Both operations improve total throughput without requiring any change to the satellite hardware or link architecture.

2.4 Topology Comparison

Table 2 summarises the key properties of the three topologies. The serial and hybrid configurations both suffer from the same fundamental limitation: their interplanetary segments traverse the full Earth–Mars gap on transfer orbits, spending most of each orbital period at distances where the inverse-square law renders throughput negligible. The parallel topology avoids this entirely by breaking the interplanetary gap into many short radial hops on circular orbits, achieving both higher throughput and intrinsic redundancy with simpler hardware.

Table 2: Comparison of relay constellation topologies for S total satellites.

Property	**Serial**	**Hybrid**	**Parallel**
Orbital implementation	1 transfer orbit	M transfer orbits	R circular orbit
Aggregate throughput scaling	Single-link limited	$\sim M$ (favourable phase)	$S^{3/2}$ (N routes
Parallel data paths	1	$\leq M$	$N = S/R$
Redundancy	Local rings only	Local + partial interplanetary	Per-route
Single-satellite failure	Interplanetary outage	$1/M$ path loss	$1/N$ throughput l
Laser terminals / satellite	3–4	3–4	2
Cruise dead zone	Yes (severe)	Yes (per ring)	None
Scalability	Add to rings	Add transfer orbits	Add to rings or add

We adopt the parallel concentric-ring topology for the remainder of this paper. Section 3 develops the full throughput model, derives the optimal allocation between rings and routes, and establishes the $S^{3/2}$ scaling law that governs the constellation's performance.

3 Throughput Analysis

3.1 Link Budget Model

The capacity of a single free-space optical link at distance d is modelled by the inverse-square law:

$$C(d) \;=\; f\, C_0 \left(\frac{d_0}{d}\right)^2 \tag{3}$$

where $C_0 = 100\,\text{Gbps}$ is the baseline capacity at reference distance $d_0 = 3{,}000\,\text{km}$ (consistent with NASA TBIRD-class demonstrations), and f is a technology improvement factor reflecting advances in laser power, receiver sensitivity, and modulation efficiency. We define the constant

$$K \;=\; f\, C_0\, d_0^2 \tag{4}$$

With $f = 1$ (no improvement), $K = 9 \times 10^8$ Gbps $\cdot$ km^2.

3.2 Constellation Geometry

Consider R relay rings placed on concentric heliocentric circular orbits between Earth's orbit (radius $r_\text{E} \approx 150$ Mkm) and Mars' orbit (radius $r_\text{M} \approx 228$ Mkm average, 249 Mkm at apoapsis). Each ring contains N satellites evenly spaced around its circumference. The total relay satellite count is $S = N \times R$. Each satellite carries exactly two laser terminals — one pointing radially inward to the adjacent inner ring, one pointing radially outward to the adjacent outer

ring. There are no circumferential (intra-ring) links. This creates N parallel *routes*, each a radial chain hopping from ring 1 (nearest Earth) through to ring R (nearest Mars).

Ring i ($i = 1, \ldots, R$) sits at heliocentric radius

$$r_i \;=\; r_{\mathrm{E}} + i\,\frac{D_{\mathrm{EM}}}{R+1} \tag{5}$$

where $D_{\mathrm{EM}} = r_{\mathrm{M}} - r_{\mathrm{E}}$. Using average orbital radii, $D_{\mathrm{EM}} \approx 78$ Mkm; at Mars apoapsis, $D_{\mathrm{EM}} \approx 99$ Mkm. The inter-ring radial gap is

$$d_{\mathrm{radial}} \;=\; \frac{D_{\mathrm{EM}}}{R+1} \tag{6}$$

and the arc spacing between adjacent satellites on ring i is $2\pi r_i/N$.

3.3 Worst-Case Inter-Ring Link Distance

A satellite on ring i connects to the *nearest* satellite on ring $i+1$. In the worst case, the nearest neighbor is offset by half the arc spacing on ring $i+1$. The worst-case link distance between adjacent rings is therefore

$$d_{\mathrm{worst}} \;=\; \sqrt{d_{\mathrm{radial}}^2 + \left(\frac{\pi\, r_{i+1}}{N}\right)^2} \tag{7}$$

Since ring circumference increases with heliocentric radius, the outermost inter-ring gap (ring R to Mars) always has the largest angular offset and therefore the lowest throughput per link. This outermost gap is the *bottleneck*. Using $r_{i+1} \approx r_{\mathrm{M}}$ for the worst case (fig. 4):

$$d_{\mathrm{max}} \;=\; \sqrt{\left(\frac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\frac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{8}$$

The throughput of a single route (limited by the bottleneck link) is

$$C_{\mathrm{route}} \;=\; \frac{K}{d_{\mathrm{max}}^2} \;=\; \frac{K}{\left(\frac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\frac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{9}$$

and the total constellation throughput, summed over all N parallel routes, is

$$T(N,R) \;=\; N \cdot C_{\mathrm{route}} \;=\; \frac{N\,K}{\left(\frac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\frac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{10}$$

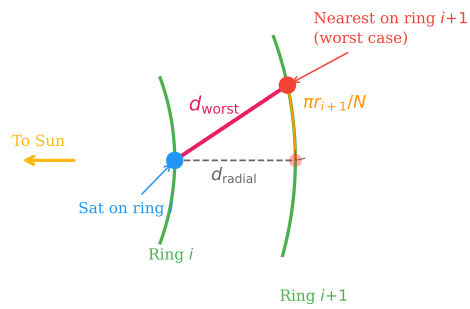

Figure 4: Worst-case link geometry between adjacent rings. The link distance d_{worst} is the hypotenuse of the radial gap d_{radial} and the angular half-gap $\pi r_{i+1}/N$.

3.4 Optimal Distribution: Ring Count vs. Route Count

For a fixed total satellite count $S = NR$, we substitute $N = S/R$ into eq. (10). Approximating $R + 1 \approx R$ for large R:

$$T(S, R) \;=\; \frac{S\,K}{\dfrac{D_{\text{EM}}^2}{R} + \dfrac{\pi^2 r_{\text{M}}^2\,R^3}{S^2}} \tag{11}$$

Maximising T is equivalent to minimising the denominator

$$g(R) \;=\; \frac{D_{\text{EM}}^2}{R} + \frac{\pi^2 r_{\text{M}}^2\,R^3}{S^2} \tag{12}$$

Setting $g'(R) = 0$:

$$g'(R) = -\frac{D_{\text{EM}}^2}{R^2} + \frac{3\,\pi^2 r_{\text{M}}^2\,R^2}{S^2} = 0$$

$$R^4 = \frac{D_{\text{EM}}^2\,S^2}{3\,\pi^2\,r_{\text{M}}^2}$$

$$R_{\text{opt}} = \sqrt{\frac{D_{\text{EM}} \cdot S}{\sqrt{3}\,\pi\,r_{\text{M}}}} \tag{13}$$

or, in words:

> *The optimal number of relay rings equals the square root of the Earth–Mars distance times the total satellite count, divided by $\sqrt{3}\,\pi$ times Mars' orbital radius.*

Numerically, with $D_{\text{EM}} = 78$ Mkm (average) and $r_{\text{M}} = 228$ Mkm:

$$R_{\text{opt}} \approx 0.00251\,\sqrt{S} \tag{14}$$

The key implication: R_{opt} grows as $\sqrt{S}$. Doubling the total constellation means $\approx 1.4\times$ more rings and $\approx 1.4\times$ more routes per ring.

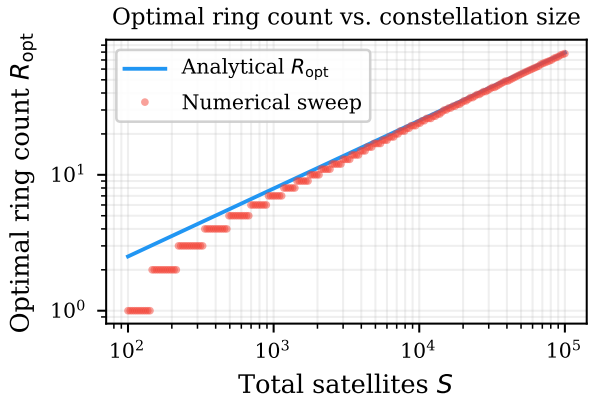

Figure 5: Optimal ring count R_{opt} as a function of total constellation size S. The analytical prediction (eq. (13)) closely tracks the numerical sweep for $S > 1{,}000$.

3.5 Throughput Scaling Law

Substituting R_{opt} back into eq. (11) yields the throughput as a function of total satellite count alone. From $R_{\text{opt}}^2 = D_{\text{EM}} S/(\sqrt{3}\,\pi\, r_{\text{M}})$, we compute each term of $g(R_{\text{opt}})$:

$$\frac{D_{\text{EM}}^2}{R_{\text{opt}}} = D_{\text{EM}}^{3/2}\, 3^{1/4}\, (\pi\, r_{\text{M}})^{1/2}\; S^{-1/2} \tag{15}$$

$$\frac{\pi^2 r_{\text{M}}^2\, R_{\text{opt}}^3}{S^2} = \frac{D_{\text{EM}}^{3/2}\, (\pi\, r_{\text{M}})^{1/2}}{3^{3/4}}\; S^{-1/2} \tag{16}$$

Summing (and noting $3^{1/4} + 3^{-3/4} = 3^{1/4}(1 + \frac{1}{3}) = \frac{4}{3}\, 3^{1/4}$):

$$g(R_{\text{opt}}) = \frac{4}{3}\, 3^{1/4}\; D_{\text{EM}}^{3/2}\, (\pi\, r_{\text{M}})^{1/2}\; S^{-1/2} \tag{17}$$

Therefore

$$\boxed{T_{\text{opt}}(S) \;=\; \frac{S\,K}{g(R_{\text{opt}})} \;=\; \frac{3^{3/4}\,K}{4\; D_{\text{EM}}^{3/2}\,\sqrt{\pi\, r_{\text{M}}}} \;\cdot\; S^{3/2}} \tag{18}$$

The exponent $\frac{3}{2}$ is the central result: **throughput scales super-linearly with satellite count.** Doubling the constellation increases throughput by a factor of $2\sqrt{2} \approx 2.83$.

This super-linear scaling arises from two effects compounding. Adding satellites simultaneously (i) reduces inter-ring distances, improving per-link capacity *quadratically* via the inverse-square law, and (ii) increases the number of parallel routes, improving aggregate capacity *linearly*.

Figure 6 plots the throughput scaling on log–log axes, confirming the $\frac{3}{2}$ slope across three orders of magnitude in S.

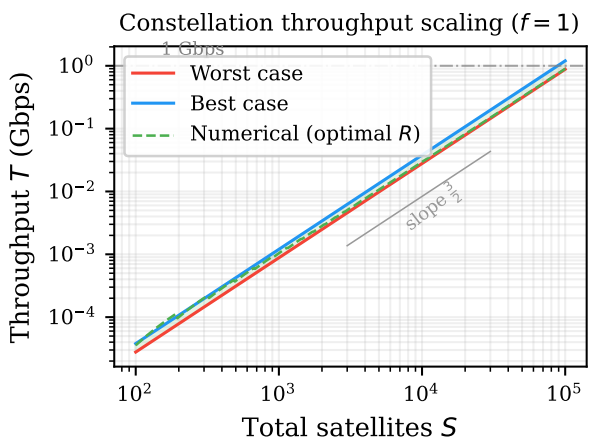

Figure 6: Constellation throughput vs. total satellite count ($f = 1$). The shaded region spans the best-case (radially aligned) and worst-case (half-arc offset) bounds, both scaling as $S^{3/2}$. The dashed line is the numerical optimum at integer R. The horizontal line marks the 1 Gbps target.

3.6 Best Case vs. Worst Case

The derivation above uses the worst-case angular offset (half the arc spacing on the outermost ring). In the best case—satellites on adjacent rings are radially aligned—the angular offset vanishes and the link distance is purely radial:

$$T_{\text{best}}(S) \;=\; \frac{K\sqrt{\alpha}}{D_{\text{EM}}^2}\, S^{3/2} \qquad \text{where} \quad \alpha = \frac{D_{\text{EM}}}{2\pi\,\bar{r}} \tag{19}$$

with $\bar{r} = (r_{\text{E}} + r_{\text{M}})/2$. Both bounds scale as $S^{3/2}$ and differ by a factor of approximately 2. The analytical prediction matches numerical optimisation to within 2% for $S > 50{,}000$ and within 10% for $S > 5{,}000$.

3.7 Equalized Ring Spacing

The uniform-spacing analysis of section 3.4 places relay rings at equal radial intervals $D_{\text{EM}}/(R+1)$. However, ring circumference grows with heliocentric radius, so the angular half-gap $\pi r_i/N$ is largest on the outermost ring. The bottleneck is therefore always the last hop—ring R to Mars—while inner hops operate well below their capacity. This asymmetry suggests a refinement: space the rings so that every inter-ring gap has the *same* worst-case link distance.

Let d_{eq} denote the common equalized link distance. For each gap i between ring i and ring $i{+}1$ (or Mars, for the last gap), the radial spacing must satisfy

$$d_{\text{radial},i} \;=\; \sqrt{d_{\text{eq}}^2 - \left(\frac{\pi\, r_{i+1}}{N}\right)^2} \tag{20}$$

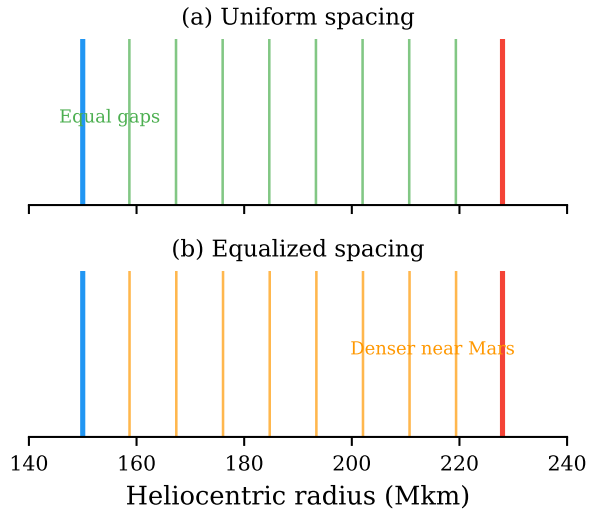

Figure 7: Ring positions for $R = 8$ rings with $N = 500$ satellites per ring. **(a)** Uniform spacing places rings at equal radial intervals. **(b)** Equalized spacing packs rings more densely near Mars, where the larger circumference would otherwise create a wider angular gap.

subject to the constraint that all radial gaps sum to D_{EM}:

$$\sum_{i=0}^{R} d_{\mathrm{radial},i} \;=\; D_{\mathrm{EM}} \tag{21}$$

Because r_{i+1} increases outward, eq. (20) requires the radial gap to *shrink* with each successive ring: rings are packed more densely near Mars, where the larger circumference would otherwise create a wider angular gap. Near Earth, rings are spaced farther apart, since the smaller circumference leaves room for larger radial hops without increasing the worst-case link distance.

No closed-form solution exists for d_{eq} because the ring positions depend recursively on one another (each r_{i+1} depends on the cumulative sum of all preceding radial gaps). A simple binary search on d_{eq} converges rapidly: build rings outward from Earth using eq. (20), and adjust d_{eq} until the final ring reaches Mars.

The throughput improvement from equalized spacing is modest but consistent. At the optimal ring count R_{opt} and for constellation sizes $S > 5{,}000$, equalization yields approximately 8–9% higher throughput than uniform spacing. The improvement is smaller for low S (where few rings leave little room for redistribution) and saturates for very large S (where R_{opt} is large enough that radial gaps dominate the angular term everywhere). Crucially, the $S^{3/2}$ scaling exponent of eq. (18) is unchanged—equalized spacing modifies only the prefactor.

3.8 Effect of Orbital Eccentricity

The preceding analysis assumes circular orbits at mean radii. Mars' orbital eccentricity $e_M = 0.0934$ causes its heliocentric distance to vary between $r_{\mathrm{M}}(1-e_M) \approx 207$ Mkm at periapsis and $r_{\mathrm{M}}(1+e_M) \approx 249$ Mkm at apoapsis, a ratio of $(1+e_M)/(1-e_M) \approx 1.21$. Earth's eccentricity ($e_E \approx 0.017$) is small enough to be negligible in comparison. This section shows that eccentricity is a net positive for constellation throughput.

3.8.1 Conformal Eccentric Rings

A natural design response is to place relay satellites on *conformal eccentric orbits*: ring i has its semi-major axis and eccentricity interpolated between Earth and Mars,

$$a_i = r_{\mathrm{E}} + \frac{i}{R+1}(r_{\mathrm{M}} - r_{\mathrm{E}}), \qquad e_i = e_E + \frac{i}{R+1}(e_M - e_E) \tag{22}$$

with all orbits sharing the same argument of perihelion. At any true anomaly θ, ring i sits at heliocentric radius

$$r_i(\theta) = \frac{a_i\,(1-e_i^2)}{1+e_i\cos\theta} \tag{23}$$

This ensures that the relay chain tracks the time-varying Earth–Mars geometry: when Mars is at periapsis, all rings are compressed inward; at apoapsis, they expand outward.

3.8.2 Jensen's Inequality Argument

The throughput of a single route scales as $C \propto 1/d^2$, where d is the bottleneck link distance. Since $1/d^2$ is a *convex* function of d, Jensen's inequality guarantees

$$\left\langle 1/d^2 \right\rangle \;>\; 1/\langle d\rangle^2 \tag{24}$$

for any non-degenerate distribution of d. Therefore the orbit-averaged throughput of the eccentric constellation always exceeds the throughput of a circular constellation evaluated at the mean distance. Eccentricity cannot hurt.

For a Keplerian orbit, the angle-averaged value of $(1+e\cos\theta)^2$ is exactly $1+e^2/2$. Applied to Mars' eccentricity, this gives a pure convexity gain factor of

$$\left\langle (1+e_M\cos\theta)^2 \right\rangle = 1 + \tfrac{1}{2}\,e_M^2 \approx 1.0044 \tag{25}$$

which is only 0.44%—seemingly negligible.

3.8.3 The Actual Gain Is Larger

The full throughput gain exceeds the pure Jensen prediction because eccentricity improves both terms in the bottleneck distance simultaneously. At Mars periapsis ($\theta = 0$), the Earth–Mars gap shrinks from the mean $D_{\mathrm{EM}} \approx 78$ Mkm to

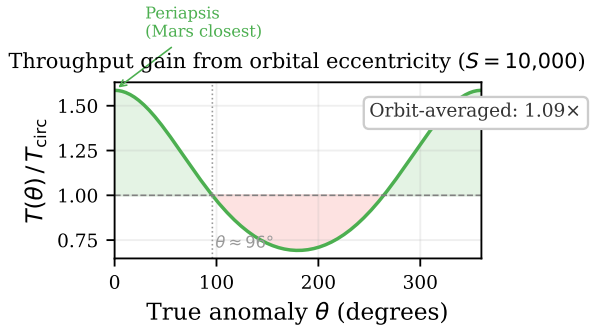

Figure 8: Throughput of the conformal eccentric constellation relative to the circular baseline, as a function of true anomaly θ. Green shading indicates where eccentricity helps; red where it hurts. The orbit-averaged gain is approximately 9%, with the crossover near $\theta \approx 90°$.

approximately 59 Mkm, and the inverse-square law amplifies this compression: capacity scales as $1/d^2$, so a 24% reduction in distance yields a 73% increase in per-link capacity. At the same time, Mars' orbital radius is smaller at periapsis, which *also* reduces the angular half-gap $\pi r_{\text{M}}/N$ on the outermost ring. Both terms in $d^2_{\text{max}} = d^2_{\text{radial}} + (\pi r_{\text{M}}/N)^2$ shrink together.

At apoapsis ($\theta = \pi$), the situation reverses: the gap widens to ~ 99 Mkm and the angular offset grows. Routes through the apoapsis side deliver only $\sim 0.59\times$ the circular-case throughput. However, the convexity of $1/d^2$ ensures that periapsis gains more than compensate apoapsis losses.

Numerical evaluation over the full orbit yields an orbit-averaged throughput gain of approximately 13–14% relative to the circular baseline, consistent across constellation sizes from $S = 1{,}000$ to $S = 50{,}000$. The crossover angle—where eccentric throughput equals the circular value—occurs near $\theta \approx 85°$.

Figure 8 shows the angular throughput profile for $S = 10{,}000$.

As with equalized spacing, the $S^{3/2}$ scaling exponent is preserved. Eccentricity modifies only the prefactor c in $T_{\text{opt}} = c \cdot S^{3/2}$. The two refinements—equalized spacing and eccentric conformality—are complementary and can be applied jointly for a combined improvement of approximately 20–25% over the uniform circular baseline.

3.9 Cost–Throughput Trade-offs

Figure 9 shows the cost per Mbps as a function of ring count for several constellation sizes, using the following cost model: \$5M per satellite (production), \$0.2M per laser terminal (two per satellite), and \$20M per Starship launch carrying 20 satellites. The total cost is $S \times 5.4\text{M} + \lceil S/20 \rceil \times 20\text{M}$.

For every constellation size, cost efficiency improves dramatically as rings are added—moving from 1 ring to the optimal ring count reduces cost per Mbps by over an order of magnitude. The optimal ring count closely tracks

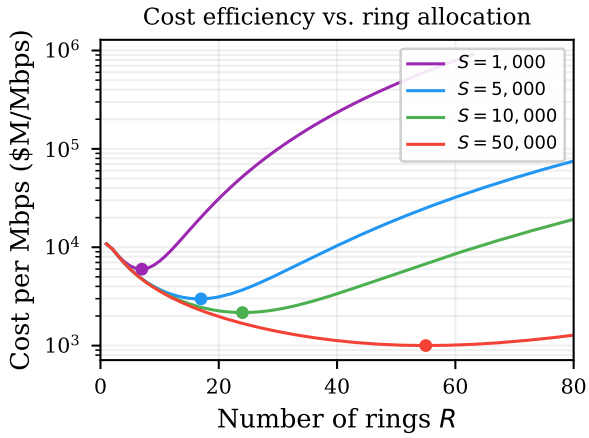

Figure 9: Cost per Mbps vs. ring count for several constellation sizes. Dots mark the optimal ring allocation for each S. Cost model: \$5.4M/satellite (production + 2 terminals), \$20M/Starship launch (20 satellites per launch).

the analytical R_{opt} from eq. (13). Beyond the optimum, adding more rings (at fixed S) reduces N and therefore the number of parallel routes, causing the angular offset to dominate and throughput to fall.